

WHAT CAN A PERSISTENCE DIAGRAM SEE?

INVARIANCE STRUCTURE AND THE DETECTABILITY OF DATA POISONING

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Motivation

- **Data poisoning** corrupts ML-based Network Intrusion Detection Systems (NIDS) by altering their training data.
- The defense must run **before training**, so clustering is unsupervised — there are no labels to use.
- Monkam et al. [1] use TDA as that filter: bytes $\rightarrow$ image $\rightarrow$ cubical persistence $\rightarrow$ clustering. **What can that lens see?**

Related Work

- **Cohen-Steiner et al. (2005) [3]**. Diagrams are stable: small perturbations give small bottleneck change. The warrant for topology as a detector.
- **Monkam, De Lucia & Bastian (2024) [1]**. The pipeline reconstructed here; reports 40–70% of poison removed.
- **Ferrara (2025) [2]**. Thm. 2 couples classification error to topological distortion — under hypotheses (linear classifiers, Vietoris–Rips, Frobenius balls) that all fail here. A complementary axis: *which points* to attack, where we ask *which algebra* registers.

Research Question

What determines a data poisoning attack’s topological detectability — and can a more sophisticated attack lead to less detectable topological signatures?

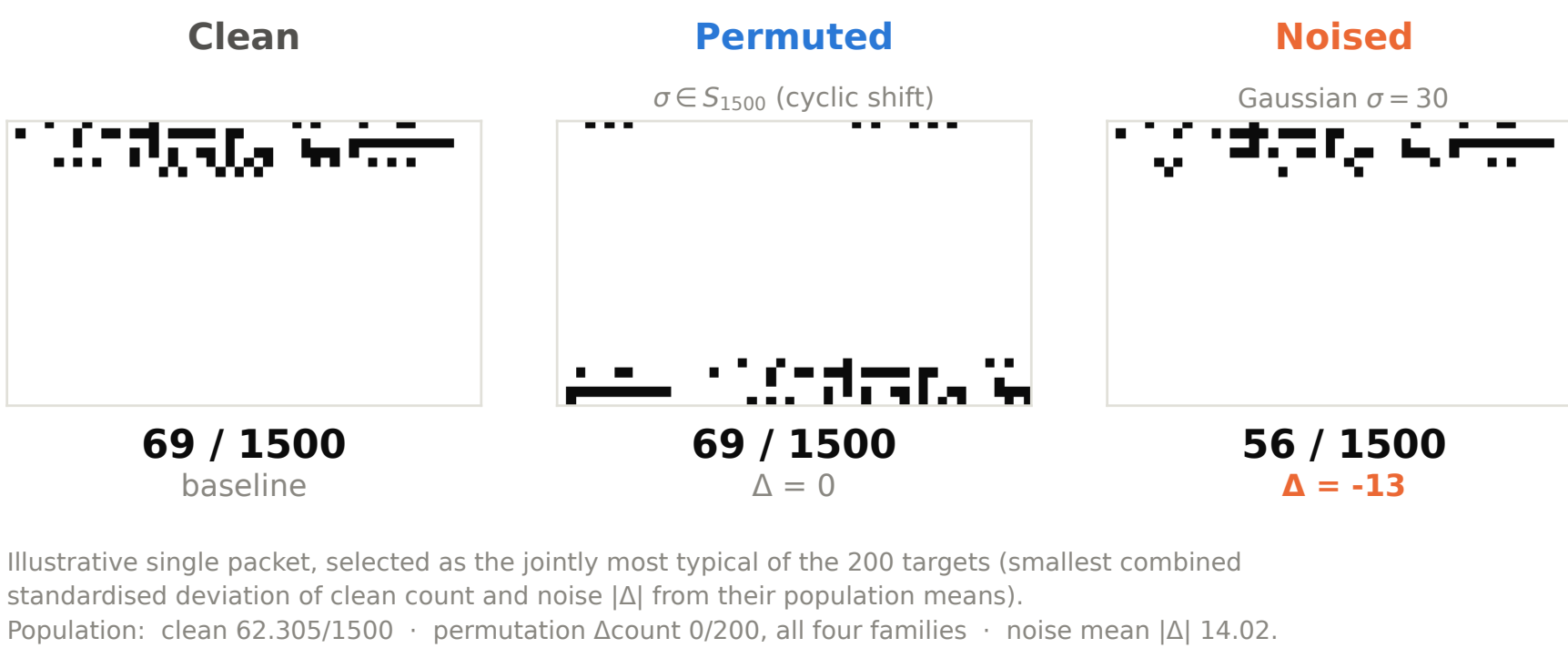
Methods I: An Invariance Argument

Let $(x \circ \sigma)_i = x_{\sigma(i)}$ for $\sigma \in S_{1500}$. Binarization b is pointwise, so it commutes with σ :

$$\underbrace{b(x \circ \sigma) = b(x) \circ \sigma}_{\text{equivariance}} \implies \underbrace{\sum_{i=1}^{1500} b((x \circ \sigma)_i)}_{\text{count is invariant } (\sigma \text{ is bijective})} = \sum_{i=1}^{1500} b(x_i)$$

- **Equivariance is not invariance.** Binarization commutes; the *count* is what is fixed. Both steps are needed, and the result holds for every σ — not just the ~ 60 transpositions applied.
- Every feature depending only on foreground count is therefore blind to permutation. Detection must flow through the **position-dependent** filtrations (Height, Radial). Invariance that is too coarse is a known hazard — the motivation for group-equivariant operators [4].
- The threshold is a fraction of the batch maximum — data-dependent, but a maximum is itself permutation-invariant. **Verified in code**, not assumed.

Permutation rearranges foreground pixels; it never changes how many
One representative packet, binarized at threshold 0.4 (fitted max_value_ = 255); foreground pixel count beneath



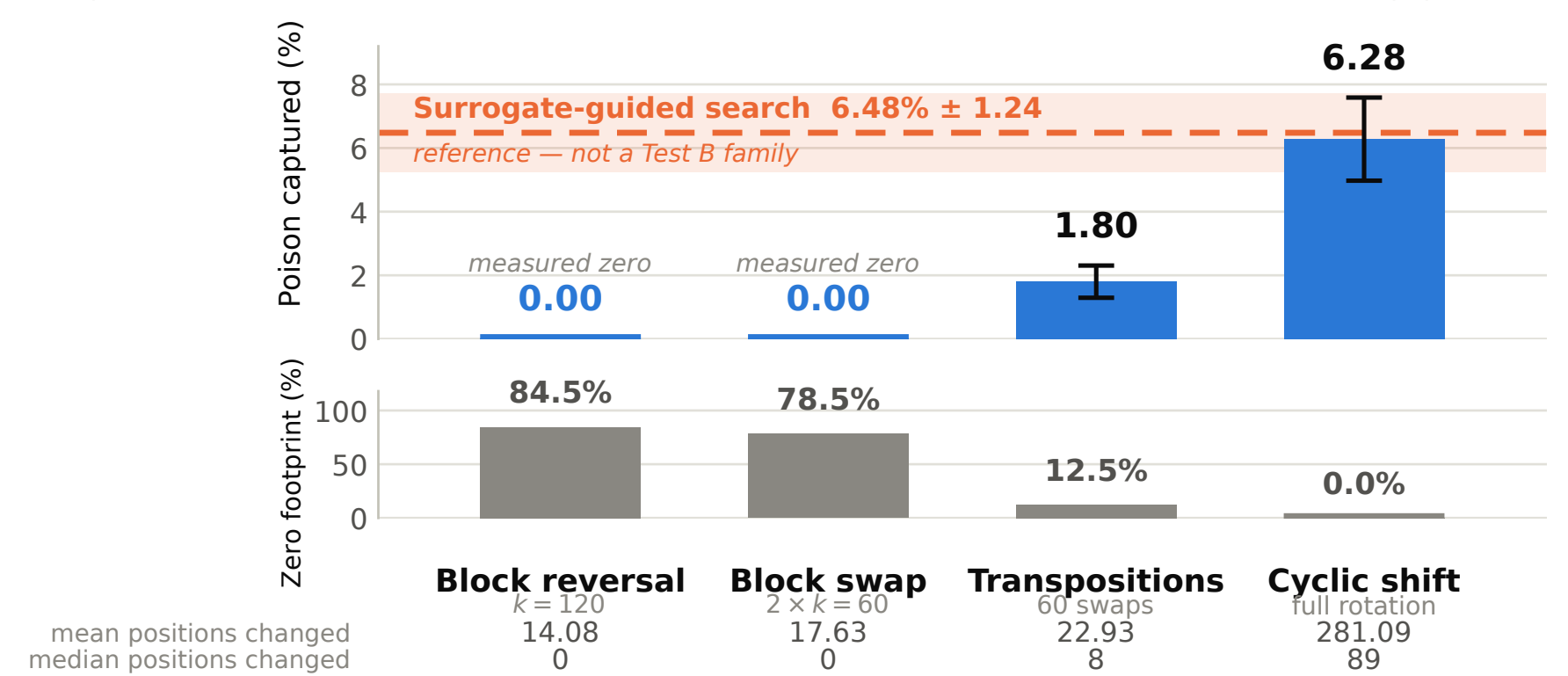
The count channel is provably insensitive to permutation; the position channel stays open.
Attenuated on the foreground-count channel — not absent.

Methods II: Design

- UNSW-NB15, 5,000 packets, 10% poison, 5 seeds. Each 1500-byte payload $\rightarrow 30 \times 50$ image; 5 filtrations $\times$ 6 vectorizations $\times$ 2 homology dimensions = 60 features (reduced homology) — the binarize–filtration–cubical pattern of Garin & Tauzin [5].
- **Capture** = fraction of the 500 appended poisoned packets assigned to a cluster containing no clean packet. Zero-tolerance: a 99%-poison cluster counts for nothing. Reported $\pm$ values are population SD across 5 seeds.
- OPTICS reported throughout. DBSCAN ≈ 0 ; MeanShift ≈ 0 but its auto-bandwidth yields coarse clusters, so it is not a discriminating cross-check. HDBSCAN is nonzero only on cyclic shift, at 3/5 seeds.
- Four multiset-preserving families preserve packet length, byte range, and byte multiset; Gaussian noise is a positive control. This is structural validity, not protocol-semantic validity.

Results: Four Families, One Proof

Detectability tracks spatial disruption, not attack sophistication
Four permutation families · UNSW-NB15 · OPTICS · 5 seeds (42/123/456/789/1024) · error bars are population SD



- **The count never moved:** 0/200 for all four families, across 1,600 checks at two thresholds — zero exceptions.
- **Identical invariance, a six-point spread in capture.** The invariance class alone does not determine detectability.
- **Random swaps are quiet for a computable reason:** foreground fraction $p \approx 0.042$, so only $2p(1-p) \approx 8\%$ of transpositions change the image. Control: noise shifts the count by $|\Delta| = 14.02$.

Discussion

- **Capture tracks spatial disruption, not sophistication:** unguided cyclic shift matches surrogate-guided search.
- **A reported negative result.** Guidance gave $+4.68 \pm 1.57$ (5/5 seeds) on UNSW-NB15 but **does not replicate** on CICIDS2017: -0.60 ± 3.03 , 2/5 seeds, per-seed $[-0.4, -2.2, +0.4, +4.2, -5.0]$ — the sign flips and the mean crosses zero. Pre-committed; not rescued.
- **The two zeroes are saturated under this metric:** for block reversal and block swap, exact feature duplicates plus OPTICS noise leave at most 0.40 and 0.32 percentage points, respectively, available for 100%-pure capture. This is not merely a clustering failure.

Conclusions

- One channel of the detector is **provably insensitive** to permutation attacks. Stability [3] bounds how far a diagram can move; it gives no lower bound — a large attack can move it *not at all*.
- **Insensitive is not undetectable:** four identically invariant families span 0%–6.3%.
- Detectability is a property of the **(attack, detector) pair** — blind spots *arising from feature-map invariances* can be read off the algebra before any experiment. The algebra does not characterise every failure mode.

Limitations. Proof: any fixed-threshold pipeline. Empirics: one reconstruction, one primary dataset, two thresholds, 5 seeds; cross-dataset replication attempted and failed. The duplicate ceilings apply only to the 100%-purity metric, not relaxed-purity sweeps.

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References

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